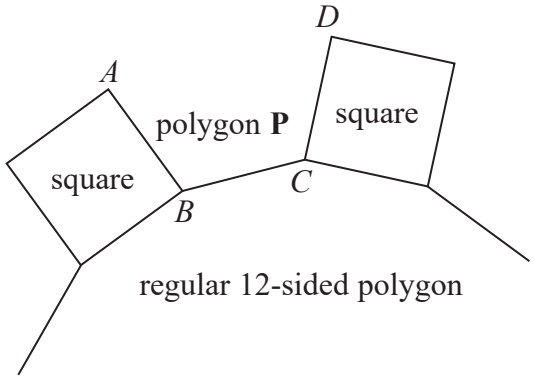
19 In the diagram, AB, BC and CD are three sides of a regular polygon **P**.



Show that polygon **P** is a hexagon.
You must show your working.

Exterior angle of the regular 12-gon = 360 ÷ 12 = 30°.

Its interior angle = 180 – 30 = 150°.

1 mark
Find a 12-gon angle

At B, angles around the point add to 360°.

Interior angle of P = 360 – 150 – 90 = 120°.

1 mark
Use the square and the 12-gon at B

1 mark
Correct angle values

Exterior angle of P = 180 – 120 = 60°.

Number of sides = 360 ÷ 60 = 6.

Therefore P is a hexagon.

1 mark
Complete proof using accurate angles

(Total for Question 19 is 4 marks)